

ABHISHEK DWIVEDI

Computational Biologist | RNA-Seq, Variant Analysis & Multi-Omics | Bioinformatics Pipelines & ML

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PROFESSIONAL SUMMARY

Bioinformatics graduate with hands-on experience across RNA-Seq, whole-genome sequencing and variant analysis, and multi-omics data integration, building reproducible Python and R pipelines and applying machine learning for genomic classification and biomarker-style feature discovery. Skilled at translating complex sequencing data into clear biological insights, figures, and reports. Brings additional depth in computational drug discovery (molecular docking, molecular dynamics, QM/MM). Comfortable working independently in high-autonomy, computationally driven research settings, with strong attention to detail, rigorous evaluation of scientific evidence, and a collaborative, curiosity-driven approach to team science.

CORE COMPETENCIES

- **Genomics & NGS:** RNA-Seq, WGS, variant analysis/calling, BLAST, sequence alignment, phylogenetic analysis, primer design; FastQC, MultiQC, Trimmomatic, HISAT2, STAR, Salmon, SAMtools, BEDTools
- **Multi-Omics & Microbial Genomics:** 16S rRNA/DADA2 amplicon profiling, AMR gene profiling (CARD), virulence factor analysis (VFDB), MLST typing, comparative genomics
- **Machine Learning & Biomarker Discovery:** Classification, regression, feature selection, PCA, Random Forest, SVM, neural networks, explainable AI (XAI)
- **Programming & Tooling:** Python (pandas, NumPy, scikit-learn, Matplotlib, Seaborn), R (DESeq2, Bioconductor, ggplot2, DADA2), SQL, C++, Linux/Bash, Conda, HPC, Git/GitHub
- **Computational Drug Discovery:** Molecular docking (AutoDock, AutoDock Vina), molecular dynamics (GROMACS, REMD, SMD), ADMET profiling, homology modelling, QM/MM modelling

SELECTED PROJECTS

Automated RNA-Seq Pipeline

- End-to-end pipeline covering QC (FastQC, MultiQC), trimming (Trimmomatic), alignment (HISAT2/STAR), quantification (Salmon), and differential expression analysis (DESeq2)

NGS Pipeline — Read Processing & Variant/Feature Calling

- Reproducible pipeline for read processing (SAMtools, BEDTools), alignment, and variant/feature calling

16S rRNA DADA2 Pipeline

- Amplicon sequencing workflow using DADA2 for taxonomic profiling and microbial community analysis

QM/MM Modelling of Protein–Ligand Systems

- Hybrid QM/MM workflow modelling electronic-level interactions at protein active sites
(github.com/ABHIVEDI11/QM-MM)

Molecular Dynamics Workflows (REMD & SMD)

- Enhanced-sampling (REMD) and steered MD (SMD) simulations for protein–ligand stability and dissociation pathways in GROMACS

Full portfolio: github.com/ABHIVEDI11

M.SC. THESIS

Antimicrobial Resistance Analysis of *Salmonella enterica* Using Whole Genome Sequencing

Jan 2026 – Jun 2026

- Compiled and analysed a panel of 67 *Salmonella enterica* genomes sourced from BV-BRC and NCBI
- Profiled antimicrobial resistance genes (CARD) and virulence factors (VFDB) across the genome panel
- Performed MLST typing and phylogenetic reconstruction using MAFFT and FastTree
- Applied machine-learning-based feature engineering and classification to AMR/virulence profiles — a workflow directly analogous to biomarker discovery from genomic data

EDUCATION

M.Sc. Bioinformatics

Aug 2024 – 2026

Central University of Himachal Pradesh | CGPA: 8.33/10.0 (Semesters 1–3; final semester results awaited)

B.Sc. Biotechnology

Jul 2019 – Jun 2022

Amity University, Lucknow Campus | CGPA: 9.02/10.0

RESEARCH & PROFESSIONAL EXPERIENCE

Machine Learning Virtual Intern

Aug 2025

CodeAlpha

- Built and evaluated supervised learning pipelines (classification & regression) on real-world datasets using scikit-learn, pandas, and NumPy

Research Contributor – Review Article

May 2021 – Jul 2022

Amity University, Lucknow Campus

- Contributed to literature review and manuscript preparation for a review article on telemedicine in COVID-19 management
- Conducted systematic literature mining across PubMed, Google Scholar, and institutional databases; performed evidence synthesis on large, heterogeneous clinical datasets

CERTIFICATIONS & TRAINING

- Computational Drug Discovery, Docking, MD & QM/MM Approaches — DE·LIFE, May 2026
- Genomic Data Science: Tools, Trends & Translation — NIPGR / BRIC India, Mar 2026
- CRISPR & Gene Editing – 3-Day Workshop — Eduvea, Jan 2026
- National Workshop on Whole Genome Sequencing & RNA-Seq — BioTecNika, Dec 2025
- CADD Master Certification – 20-Day Program — BioTecNika, Jun 2025
- Python Bootcamp – Zero to Hero — Udemy / Pierian Training, Jul 2025
- Mastering R for Bioinformatics — Dr. Omics Labs, May 2025
- Bioinformatics Methods & Tools Master Program — BioTecNika, Nov 2024